

HTML Basics

QUICK DEFINITION

HTML (HyperText Markup Language) is the standard language used to structure content on the web using nested elements.

● STRUCTURE

Elements

Content is built from nested tags like headings and paragraphs.

● RENDER

Browser

The browser reads the tags and displays content accordingly.

● MEANING

Semantics

Tags label content: this is a heading, this is a link, etc.



index.html

EXAMPLE

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>My Page</title>
5   </head>
6   <body>
7     <h1>Welcome</h1>
8     <p>This is a paragraph.</p>
9   </body>
10 </html>
```

HTML Elements

QUICK DEFINITION

An element is a start tag, its content, and a matching end tag. Some elements are self-closing and hold no content.

● PAIRED

Container

Most tags wrap content between an opening and closing tag.

● EMPTY

Self-closing

Tags like `<br>` and `<img>` have no closing tag or content.

● NESTED

Hierarchy

Elements can contain other elements, forming a tree.

```
1 <p>A paragraph with a </  
2   <br>  
3   line break inside it.</p>  
4  
5   
6  
7 <div>  
8   <span>nested text</span>  
9 </div>
```

elements.html

EXAMPLE

HTML Attributes

QUICK DEFINITION

Attributes add extra information to an element and always appear in the opening tag as name="value" pairs.



ID

Unique

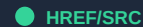
id gives one element a unique identifier on the page.



CLASS

Reusable

class groups elements to target with CSS or JS.



HREF/SRC

Links data

href and src point to a linked resource or file.



attributes.html

EXAMPLE

```
1 <a href="https://example.com">
2   Visit site
3 </a>
4
5 
6
7 <p class="intro" id="lead">
8   Styled paragraph</p>
```

Headings & Paragraphs

QUICK DEFINITION

Headings (h1–h6) define a hierarchy of importance, and paragraphs (p) group blocks of readable text content.

● H1–H6

Hierarchy

h1 is the most important; h6 the least. Use in order.

● PARAGRAPH

Body text

The <p> tag wraps a standalone block of text.

● EMPHASIS

Inline style

 and add bold and italic meaning.



text.html

EXAMPLE

```
1 <h1>Main Title</h1>
2 <h2>Section Heading</h2>
3
4 <p>
5   This text is <strong>important</strong>
6   and this is <em>emphasized</em>.
7 </p>
```

Links & Images

QUICK DEFINITION

Links (a) connect pages together, and images (img) embed visual content — both rely on attributes to work.

● A TAG

Hyperlink

href sets the destination URL; target="_blank" opens a tab.

● IMG TAG

Picture

src points to the file; alt describes it for accessibility.

● RELATIVE

Local paths

Paths like "images/pic.jpg" link to files in your project.

```
1 <a href="page2.html">Next page</a>
2
3 <a href="https://ex.com" target="_blank">
4   Open in new tab</a>
5
6 
```

media.html

EXAMPLE

Lists & Tables

QUICK DEFINITION

Lists (ul/ol/li) group related items, and tables (table/tr/td) organize data into rows and columns.

● UL/OL

Lists

ul is unordered (bullets); ol is ordered (numbers).

● LI

List item

Each entry in a list is wrapped in its own tag.

● TABLE

Grid data

tr defines a row; td a cell; th a header cell.

```
1 <ul>
2   <li>Coffee</li>
3   <li>Tea</li>
4 </ul>
5
6 <table>
7   <tr><th>Name</th></tr>
8   <tr><td>Ada</td></tr>
9 </table>
```

lists.html

EXAMPLE

HTML Forms

QUICK DEFINITION

Forms (form) collect user input through fields like text boxes, checkboxes, and buttons, then submit the data.

● FORM

Container

Wraps all input fields and defines where data is sent.

● INPUT

Field types

type="text/email/password" changes the input's behavior.

● LABEL

Accessibility

label ties readable text to a specific input field.

```
1 <form action="/submit">
2   <label>Name:</label>
3   <input type="text" name="user">
4
5   <input type="submit" value="Send">
6 </form>
```

form.html

EXAMPLE

Inline Styles & Classes

QUICK DEFINITION

The style attribute applies CSS directly to one element, while class/id link elements to rules in a stylesheet.

● STYLE

Inline CSS

style="color:red;" affects only that single element.

● CLASS

Reusable rule

Shared styling applied to many elements via a class name.

● STYLE TAG

Page-wide

A <style> block in <head> defines rules for the whole page.

```
1 <p style="color:blue;">
2   Blue text</p>
3
4 <style>
5   .highlight { background: yellow; }
6 </style>
7
8 <p class="highlight">Highlighted</p>
```

style.html

EXAMPLE